

Medha Mishra

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EDUCATION

Netaji Subhas University of Technology

B.Tech. in Computer Science and Engineering; CGPA: 7.85/10

Delhi, India

Aug. 2022 – May 2026

Amity International School, Mayur Vihar

Delhi, India

Senior Secondary (CBSE, Class XII); 95.6%

2021 – 2022

Delhi Public School, Indirapuram

Ghaziabad, India

Matriculation (CBSE, Class X); 98.4%

2019 – 2020

EXPERIENCE

Business Technology Solutions Associate

Jan 2026 – Present

ZS Associates

Gurugram, India

- Building full-stack features for **Kural**, a flagship data-driven decision platform supporting revenue streams exceeding \$100M, using React.js, FastAPI, and AWS
- Optimizing backend APIs and data pipelines to improve response times and reliability for enterprise clients
- Collaborating within a cross-functional data engineering team to ship and integrate scalable production features

Software Development Engineer Intern

May 2025 – Jul. 2025

TechAhead

Noida, India

- Engineered interactive SaaS product features with JavaScript, CSS, Redux, and EmailJS, improving UI responsiveness and workflow automation
- Built reusable components and state-management patterns in agile sprints, accelerating feature rollout and reducing manual effort across product surfaces

PROJECTS

Federated Learning for Alzheimer's Classification | *PyTorch, ViT, ResNet, DP-SGD* | [GitHub Link](#) 2025 – 2026

- Built a hybrid CNN-Transformer model (ResNet-18 + ViT-Base) fusing local and global MRI features for 4-class Alzheimer's staging, reaching 89.5% accuracy, a 5-point gain over a ResNet-only baseline
- Engineered a privacy-preserving federated pipeline with FedProx and Dirichlet non-IID partitioning, recovering 4.7% of the accuracy lost by FedAvg under simulated inter-hospital data heterogeneity
- Integrated DP-SGD via Opacus for formal (ϵ, δ) -differential privacy at only ~3% accuracy cost, enabling HIPAA/GDPR-compliant multi-site training

SignBridge – Sign Language Game | *Python, TensorFlow, MediaPipe, OpenCV* | [GitHub Link](#)

Jan – Apr. 2025

- Built a gamified sign-language learning platform combining computer vision and deep learning, achieving 95%+ model accuracy
- Designed a real-time gesture-recognition pipeline with MediaPipe and OpenCV, integrated with React.js via REST APIs; added MongoDB-backed progress tracking and adaptive difficulty

Voice-Enabled Health Assistant | *React.js, FastAPI, OpenAI, MongoDB* | [GitHub Link](#)

Oct. 2025

- Built a scalable voice-based AI healthcare assistant handling pauses and disfluencies in natural speech; secured an internship offer from Noise
- Integrated OpenAI APIs with a FastAPI backend and MongoDB for low-latency conversational responses and session persistence

PUBLICATIONS

Smart Cities & Sustainable Infrastructure: AI at Work | *Springer Nature* | [Publication](#)

Published May 2026

- Proposed an AI-driven smart-traffic management framework using Graph Neural Networks and Reinforcement Learning for adaptive signal optimization, demonstrating up to 40% reduction in vehicle queue length in simulation (Scopus-indexed)

ACHIEVEMENTS

Nokia Ideathon 2024 – Nationwide Finalist (Top 10) | *Python, MATLAB, REST APIs*

May – Jul. 2024

- Explored AI-assisted 6G communication for beyond line-of-sight transmission in coastal regions using atmospheric ducting modeling

Myntra HackerRamp 2024 – Top 1200 of 30,000+ Teams | *Python, ML, HTML/CSS*

Jun. 2024

- Built MyntraOnBudget, an ML-powered recommender using cosine-similarity models targeted at budget-conscious Gen-Z users

TECHNICAL SKILLS

Languages: Python, C, C++, JavaScript, SQL, R, HTML/CSS

Frameworks & Libraries: React.js, Node.js, FastAPI, Flask, Redux, PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas

ML & Data: Computer Vision, Vision Transformers, Federated Learning, Differential Privacy, GNNs, Power BI, Tableau

Cloud & Databases: AWS (EC2, S3, IAM, RDS, DynamoDB), MySQL, MongoDB

Developer Tools: Git & GitHub, REST APIs, Streamlit, Tailwind CSS, Linux

VOLUNTEER EXPERIENCE

Google Developers Student Club, NSUT | *Senior Operations Manager*

- Managed on-site and remote events including interviews, orientations, and workshops, impacting 1000+ students

MARS Mindshift Training Hub | *Technical Lead*

- Led technical initiatives, workshops, and student mentorship programs across the hub's community of learners